BIT 201 Group assignment(1)

اعداد/ AAKROSHAN CHAUDHARY -

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Assignment Cover Sheet

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Module/Subject Information		**Office Acknowledgement**
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Lecturer/Tutor/Facilitator	Nirajan Poudel	
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Signature: Aakroshan Chaudhary	Date: 15, Feb 2026
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Feedback/Comments*	
Main Strengths	
Main Weaknesses	

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<tr><td>Student’s signature:Aakroshan Chaudhary</td></tr>
<tr><td></td><td>Date:2/15/2026</td></tr>
</table>

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1)A soft and hard copy of the assignment shall be submitted.
2)The signed copy of the assignment cover sheet shall be retained by the marker.
3) If the Turnitin report is required, students have to submit it with the assignment. The assignment shall only be marked upon the submission of the Turnitin report.

*Use additional sheets if required.

ACKNOWLEDGMENT

We would first and foremost like to sincerely thank our lecturer for the ongoing direction, insightful criticism, and scholarly assistance that we received during this module, BIT201 System Architecture and Design.

This project has been a tremendous collaborative learning experience that has enabled us to apply theoretical Object-Oriented Analysis and Design (OOAD) concepts to a real-world social problem in Nepal. We also want to thank our team members, Aakroshan Chaudhary, Aaditya Chaudhary, and Buddhiman B.K.

Lastly, we would like to express our gratitude to our families for their unwavering support throughout the writing of this report and to our peers for their helpful conversations.

Abstract

The number of senior citizens in Nepal is growing as a result of the country's demographic shift, which is driving up demand for trustworthy home-based assistance. The architectural design and system analysis of SeniorHELP, an information system created to close the gap between senior citizens in need of help and volunteers ready to offer it, are described in this report.

The team which included Aakroshan Chaudhary, Aaditya Chaudhary, and Buddhiman B.K. identified important functional requirements like user registration, service request management, and feedback mechanisms using a methodical architectural approach. Essential UML artefacts such as Activity Diagrams, Use Case Diagrams, Analysis Class Diagrams, and System Sequence Diagrams (SSD) are used to model the system. With the ultimate goal of improving senior citizens' quality of life through technology-driven community support, the suggested architecture guarantees a safe and effective platform for managing volunteer resources

Contents

1. Introduction

1.1 Background

In Nepal, the population is getting older. The percentage of elderly people is rising as a result of improved healthcare and falling fertility rates. Senior citizens were traditionally cared for in joint family structures, but as a result of urbanisation and the migration of young people into the workforce, many elderly people are now living alone without immediate physical assistance with everyday tasks.

1.2 Problem Statement

Many elderly people in Nepal's cities struggle to find friends or complete everyday tasks. Although a large number of people are willing to volunteer, there isn't a centralised, reliable platform to organise these activities. In order to solve this, SeniorHELP offers a validated online space where resources and needs can coexist in safety.

1.3 Project Objectives

The main objectives of the SeniorHELP system are:

- To offer a secure environment for senior citizens to request home-based volunteer services.
- To enable authenticated volunteers to browse and accept service requests in their area of operation.
- To enable a transparent feedback mechanism for quality service and safety.

1.4 Scope of the Report and Individual Responsibilities

This report covers the analysis and design phase of the SeniorHELP system. To ensure individual responsibility and to meet the requirements of the individual component of the assignment, the tasks were allocated as follows:

- Aakroshan Chaudhary: In charge of the Architecture Vision and the individual design of the Use Case 2 (Request Home Volunteer), including its Expanded Use Case, SSD, and Operation Contracts.

- Aaditya Chaudhary: In charge of the Activity Diagram and the individual design of the Use Case 3 (Accept Volunteer Request), including its Expanded Use Case, SSD, and Operation Contracts.

- Buddhiman B.K.: In charge of the Analysis Class Diagram and the individual design of the Use Case 4 (Provide Feedback), including its Expanded Use Case, SSD, and Operation Contracts.

Task1: ARCHITECTURE VISION

1.1. Problem Statement

Nepal is facing a swiftly growing elderly population, resulting in a notable demographic transition that poses social and healthcare difficulties. Numerous seniors reside by themselves or with relatives who are often busy with work or other duties, resulting in social isolation, feelings of loneliness, and unfulfilled non-medical needs like companionship, help with reading, and light housekeeping tasks. At present, there is no organized, trustworthy, and safe platform that links seniors with volunteers able to deliver these vital services. Families have difficulty locating reliable assistance, while volunteers do not have a unified system to discover and engage in tasks that align with their skills and schedules. As a result, a definitive requirement exists for a system that not only aids in connecting seniors with volunteers but also guarantees safety, responsibility, and a premium caregiving experience

To close this gap, the SeniorHELP system offers:

- **Veted User Management**: To build community confidence, a secure registration process for both Requestors and Service Providers.
- **Task Specificity**: A method for requesters to specify certain activities, such as tidying the home, playing games, or reading books.
- **Dynamic Matching**: An interactive dashboard enabling volunteers to browse and select tasks according to their availability and expertise.
- **Quality Governance**: An assessment and feedback framework to ensure responsibility and superior care

1.3 Stakeholders' Key Concerns

The SeniorHELP system includes various stakeholders, each having unique roles and priorities. Grasping their worries is essential to guarantee the system fulfills user expectations, upholds trust, and functions efficiently. The main stakeholders and their issues are outlined in the table below:

Stakeholder	Role	Key Concerns
Requestors (Family members)	Individuals seeking assistance for their elderly relatives	- Simple and user-friendly process for submitting requests - Dependability and credibility of designated volunteers - Tracking the status of requests in real time - Adaptability in defining types of activities and timetables
Service Providers (Volunteers)	Individuals providing home-based support	- Precise outline of responsibilities and anticipated outcomes - Easy method for exploring and approving requests - Capability to sort tasks based on location, schedule, or type of activity. - Just evaluation and acknowledgment for the services rendered
Senior Citizens (Recipients)	Elderly individuals receiving support	- Protection and assurance while conducting home visits

		- Considerate and polite communication - Interactive experiences that fulfill their social and emotional requirements. - Uniformity in volunteer roles
System Administrators	Staff managing the SeniorHELP platform	- Ensuring the privacy and security of data - Maintaining system availability and dependability - Observing user actions and addressing problems - Overseeing user verification and authentication processes

Explanation:

- Requestors require a system that streamlines the support request process while ensuring the designated volunteers are reliable and attentive.
- Volunteers need an efficient and structured method to choose assignments that align with their abilities, time, and interests.
- Elders value safety, comfort, and significant involvement since they are the main users of services.

- System Administrators ensure the platform's overall integrity and smooth functioning, making certain that the system remains secure, operational, and dependable

Task2: Business Process Model (Activity Diagram)

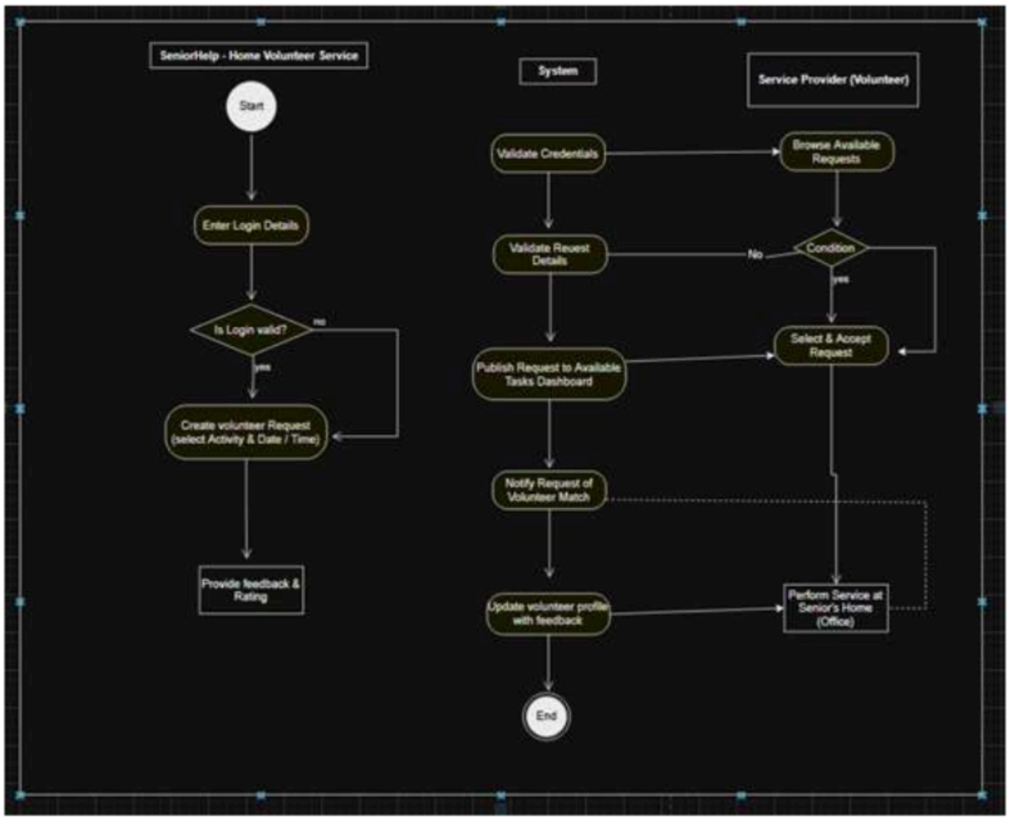


Figure 1: SeniorHelp Service Fulfillment Activity Diagram

Diagram Analysis:

The SeniorHelp business process begins with **Login Details entry and validation**. Once authenticated, the **Requestor** creates a volunteer request (selecting activity, date, and time).

On the **System** side, the request is validated and published to a dashboard where the **Service Provider (Volunteer)** can browse and accept requests based on a condition. The transition from digital interaction to offline service is marked by the **"Perform Service at Senior's Home"** action. The process concludes with the **Requestor providing feedback and rating**, which the system uses to update the volunteer's profile.

Task3:Use Case Diagram

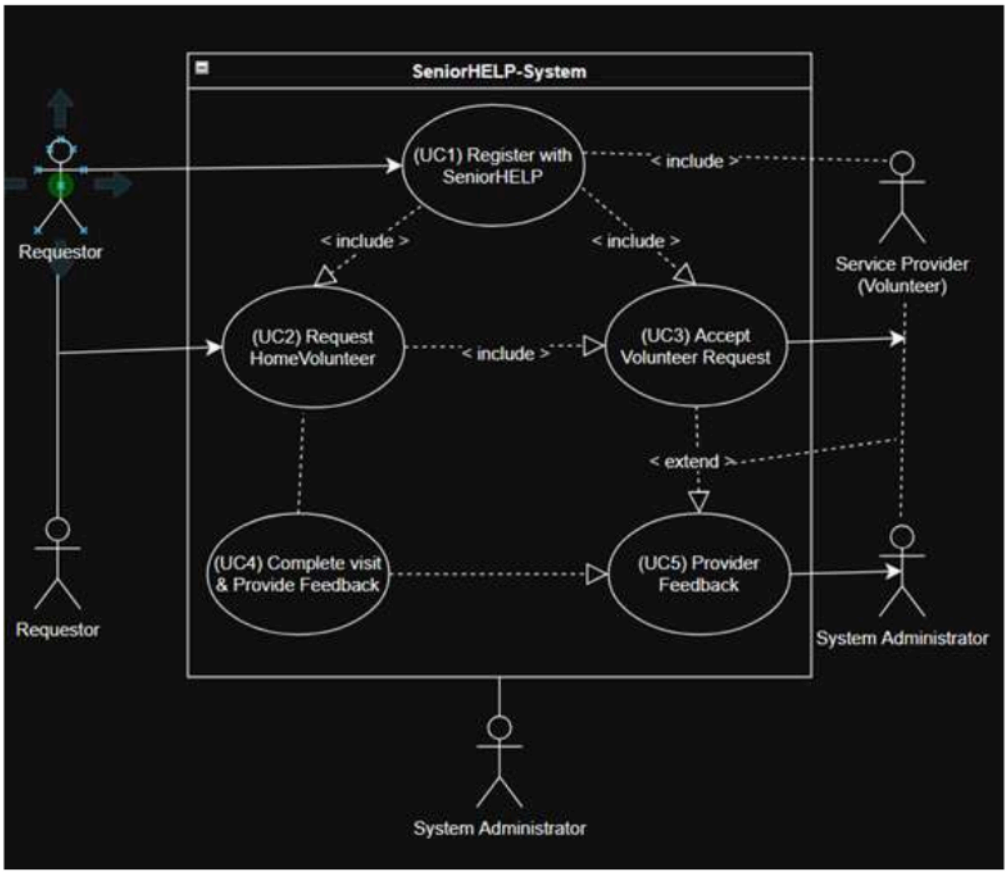


Figure 2: SeniorHELP Use Case Diagram (Home Volunteer Service)

Diagram Analysis:

The "SeniorHELP-System" boundary's functional scope is specified in the Use Case Diagram. The Requestor, Service Provider, and System Administrator are the three actors that are identified. While specific goals like "Request Home Volunteer" are limited to the Requestor role, key use cases like "Register with SeniorHELP" are shared among primary actors. All functional development is based on this model.

Task4: High-Level Use Case Descriptions

The SeniorHELP system offers vital services linking older adults with volunteers who can support them in home-related tasks. These services are structured into overarching use cases that outline the main functions of the system and its interactions with various user types. Every use case specifies the primary actor, the objective of the interaction, and a concise overview of the functionality. Collectively, these elevated use cases offer a concise summary of how the system aids requestors, volunteers, and administrators in providing and overseeing home volunteer services

Use Case Table

UCID	Use Case Name	Primary Actor	Goal	Brief Description
UC1	Register User	Requestor/ Provider	Access System	User provide information to create a secure account.
UC2	Request Volunteer	Requestor	Post Task	Indicates the type of activity (Reading / Cleaning) and duration

UC3	Select Request	Service Provider	Commit to Task	A volunteer surveys the pool and takes on a particular task
UC4	Feedback/ Payment	Requestor	Evaluate Service	The Requestor evaluates the fulfillment of the volunteer logs

Task5:Expanded Use Case – Request Home Volunteer

UC1: Register User (Group Component)

Role: Requestor / Service provider

Stakeholders & Interests:

- User: You may have to register for your app, so ensure the process is simple and secure.
- System Administrator: When your application requires trust and security, accurate and verifiable user information is essential.
- Conditions that must be met prior to proceeding.

Preconditions:

- The user accesses the SeniorHELP web/app interface by logging in..
- The user has not logged in.

Postconditions:

- ·A new user instance will be created by the system.
- A unique-profile User-ID generated by the system is assigned to the user.
- The function of the user (Requester or Volunteer)
- Guide the user to their dashboard.

Main Success Scenario:

- User selects "Register" on the main page.
- The system subsequently asks the user to select a role (Senior/Caregiver or Volunteer).
- The registration form is shown by the system.
- User provides Name, Phone Number, Mailing address, Emergency contact, and Password.
- The system verifies that:
- The phone number is not the same.
- The location is within the service's coverage zone.
- The system securely saves the user's credentials.
- User profile is created by the system.
- System directs user to their dashboard.

UC2: Request Home Volunteer (Aakroshan Chaudhary – Individual Task)

Primary actor: Senior / Caregiver Side(Core) - Click and drop target is Primary Actor Description: 1.

Preconditions:

- The requester is signed in and validated within the system.

Postconditions:

- The Request object is newly created.
- The status of the request is "Available."
- The application is submitted into the Volunteer Job Pool.

Main Success Scenario:

- The requester selects "New Request" from the dashboard.
- Types of activities shown by the system include Reading, Home Cleaning, and Games.
- The requester chooses an activity.
- Requester inputs preferred date and time slot.
- The party making the request includes any specific instructions, if there are any.
- System verifies input information.
- System determines projected time.
- System verifies request information.
- System uploads it to Volunteer Job Pool.

UC3: Accept Volunteer Request (Aaditya Chaudhary – Individual Task)

Primary Actor: Service Provider (Volunteer)

Preconditions:

- Volunteer has been confirmed.
- You are a volunteer, and your account status is "Approved."

Postconditions:

- Change request status from "Available" to "Assigned."
- The Request is linked to the Volunteer.
- The requester has been notified.

Main Success Scenario:

- Volunteer accesses the system.
- Volunteer moves to "Available Tasks."
- The system shows task listings based on location.
- Options for filtering volunteer tasks based on the type of activity.

- Volunteers select a commitment to view the complete information.
- Volunteer selects "Accept Request."
- The Volunteer is assigned and equipped for this Request.
- System changes the status request to "Assigned".
- The system sends a message to the Requestor.

UC4: Provide Feedback and Finalize (Buddhiman b.k – Individual Task)

Primary Actor: Requestor

Preconditions:

- The visit by the volunteers has concluded.
- The status of the request is "Assigned."

Postconditions

- Change the status to "Completed."
- Volunteer assessment has been revised.
- Comments are saved in the system database.

Main Success Scenario:

- System alerts Requestor to complete the request.
- Requestor chooses "Close"
- System presents Feedback Form (Rating 1-5 stars along with comments).
- The requester provides a rating and feedback.
- System verifies feedback
- Updates to the system affect the Volunteer's reputation score.
- The system stores the request.

Task6: Analysis Class Diagram

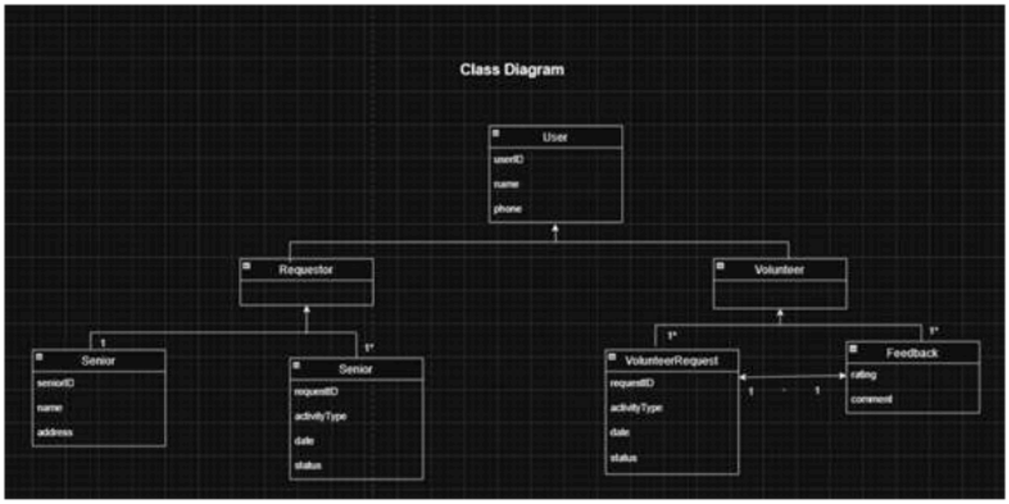


Figure 3: Class Diagram for for SeniorHELP System

6.1 Diagram Overview

The Analysis Class Diagram (Figure 3) provides a static representation of the conceptual framework of the SeniorHELP system domain. This represents a key difference from a design-level diagram, which focuses on the conceptual entities and their characteristics, as well as the logical connections among these data structures

6.2 Key Class Definitions & Attributes

The system is structured around several essential classes for managing user roles and service logistics:

- User (Parent Class): This serves as the parent class for every registered individual. It possesses characteristics such as userID, name, phone, and more.

- Requestor and Volunteer (Sub-classes): These categories will derive from the User class. The Requestor will be the family member or elder submitting the request, while the Volunteer will be the service provider.

- Senior: Includes detailed information about the elderly patient receiving support, such as their senior ID, name, and physical address.

- VolunteerRequest: This is the component that facilitates the service. It monitors significant logistics:

- requestID: A distinct identifier for the transaction.

- activityType: This field classifies the service, such as Reading, Home Cleaning, etc., as needed by Option 1.

- date: The planned time for the appointment.

- status: Represents the condition of the request, including "Available", "Assigned", and "Completed".

- Feedback: Linked to the fulfillment of a request to uphold quality standards. It captures a numerical score and a written remark from the Requestor regarding the Volunteer's performance.

6.3 Relationships and Multiplicity

The chart outlines the guidelines for how these entities engage with one another:

- Generalization (Inheritance): A continuous line with an open arrow indicates that Requestor and Volunteer are types of User and inherit all of its fundamental properties.

- Relationship: * Requestor to VolunteerRequest A single requestor can submit multiple requests for volunteers over various time periods; thus, it represents a one-to-many relationship.
- VolunteerRequest to Feedback: A one-to-one connection will guarantee that feedback is evaluated once for each completed request to uphold data integrity.
- Volunteer to VolunteerRequest: A 1 to * relationship signifies that a volunteer may accept and be assigned various requests during their service.

6.4 Technical Just

This framework is essential for ensuring the system can fulfill the needs of Option 1 (Home Volunteers). The separation of the 'Senior' (the recipient) from the 'Requestor' (the initiator) enables the system to represent a genuine scenario in which a child arranges a 'Reading' or 'Cleaning' session for an elderly person. Incorporating the 'activityType' attribute within the VolunteerRequest object enables the system to show tasks to the volunteer based on the task type, interest, or skill set

Task7: System Sequence Diagram (SSD)

Use Case 1: Register with SeniorHELP

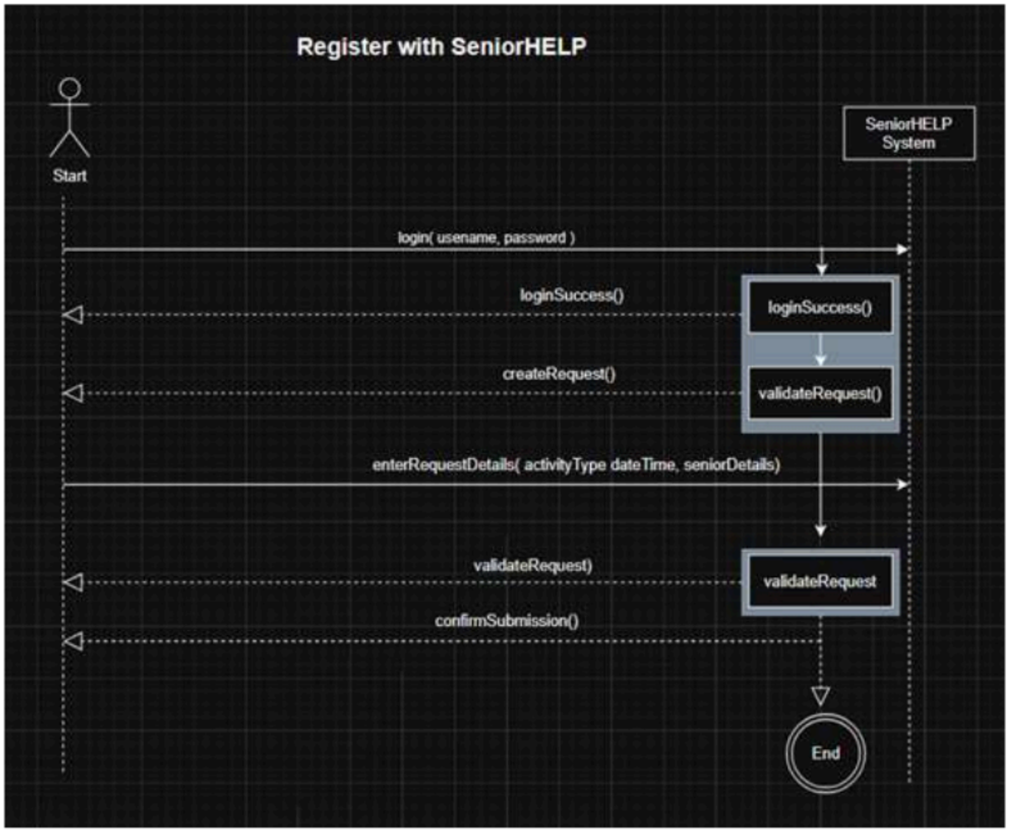


Figure: Register with SeniorHELP(SSD)

Explanation:

This Diagram displays the necessary interaction steps required for a new user to sign up and utilize the SeniorHELP system.

- Actor: Service Provider or Requestor.
- System (Black Box): SeniorHELP app interface.
- Key Interactions: User transmits the registerUser() command along with profile information.
- Furthermore, it performs internal validation, such as confirming that mobile numbers are distinct.
- The system subsequently sends a registrationConfirmation to the user, signifying that the account is active.

Use Case 2: Request Home Volunteer(**AAKROSHAN CHAUDHARY(E2300551)** – Individual Task)

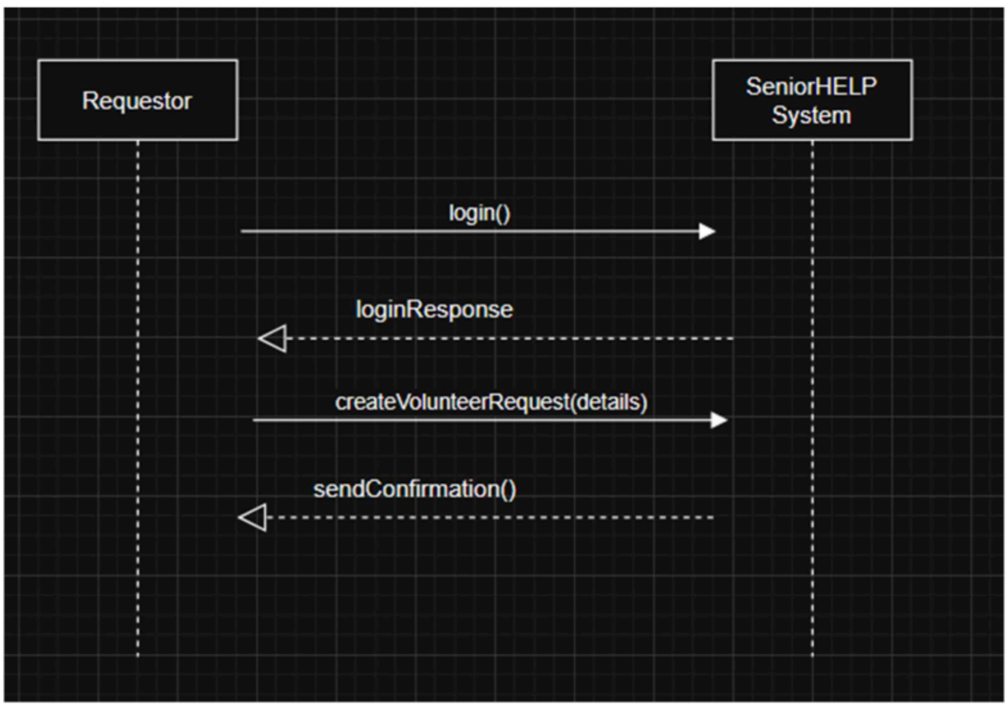


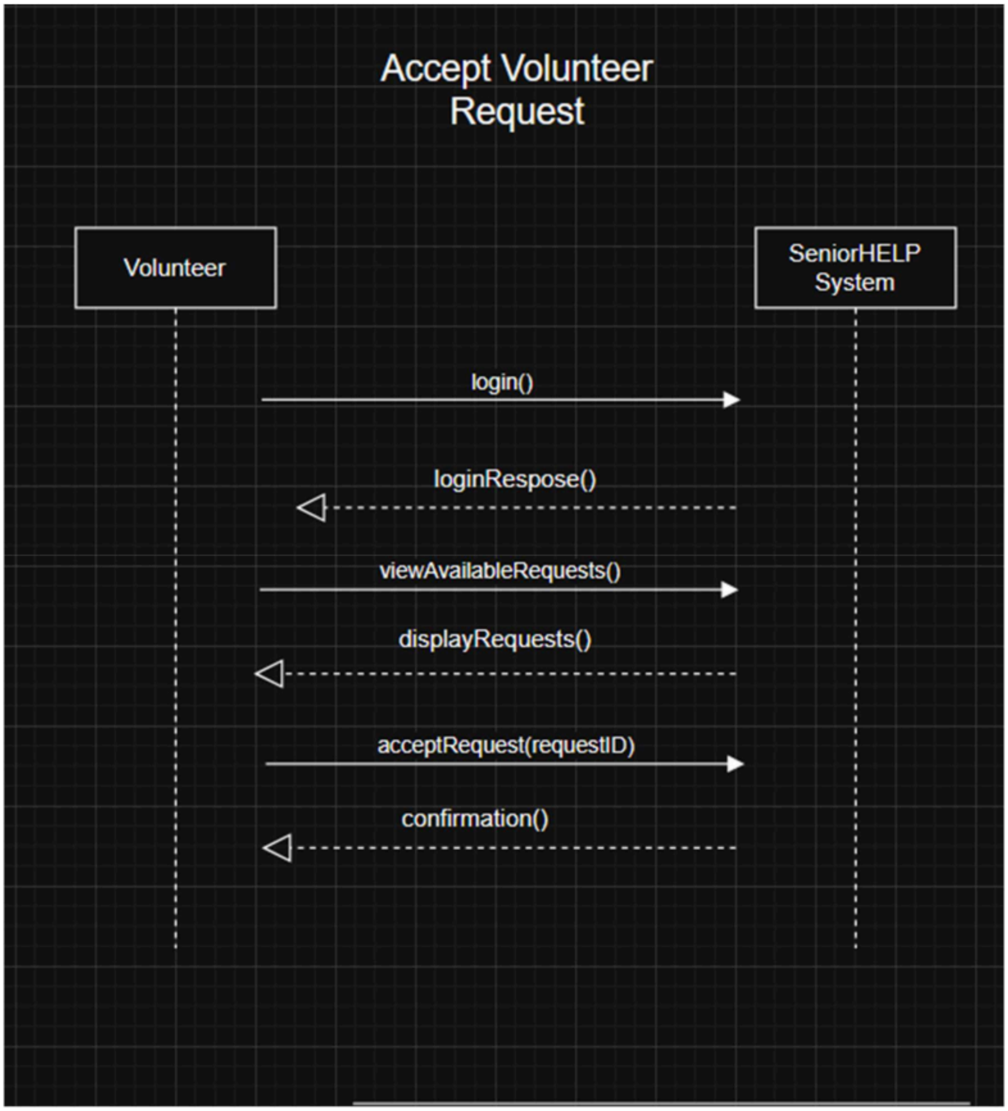
Figure: Request Home Volunteer (SSD)

Technical Description:

Since this SSD is the primary contact for Member 1's task, it will focus on the "Requesting" phase of Option 1 (Home Volunteers).

- Actor: Inquirer
- Core Messages: login(credentials): Validates the session.
- makeRequest(activityType, time, date): The Requestor submits the designated activity, such as "Reading" or "Cleaning."
- Internal Loop: The diagram illustrates that the system is presently carrying out an internal callback function, addToAvailableRequests(). This step is crucial because, during this stage, the data will be transferred to be displayed on the public dashboard for volunteers to view.
- Final Response: A confirmation message is received once the request is active.

Use Case 3: Accept Volunteer Request**(AADITYA CHAUDHARY(E2300548) - INDIVIDUAL TASK)**



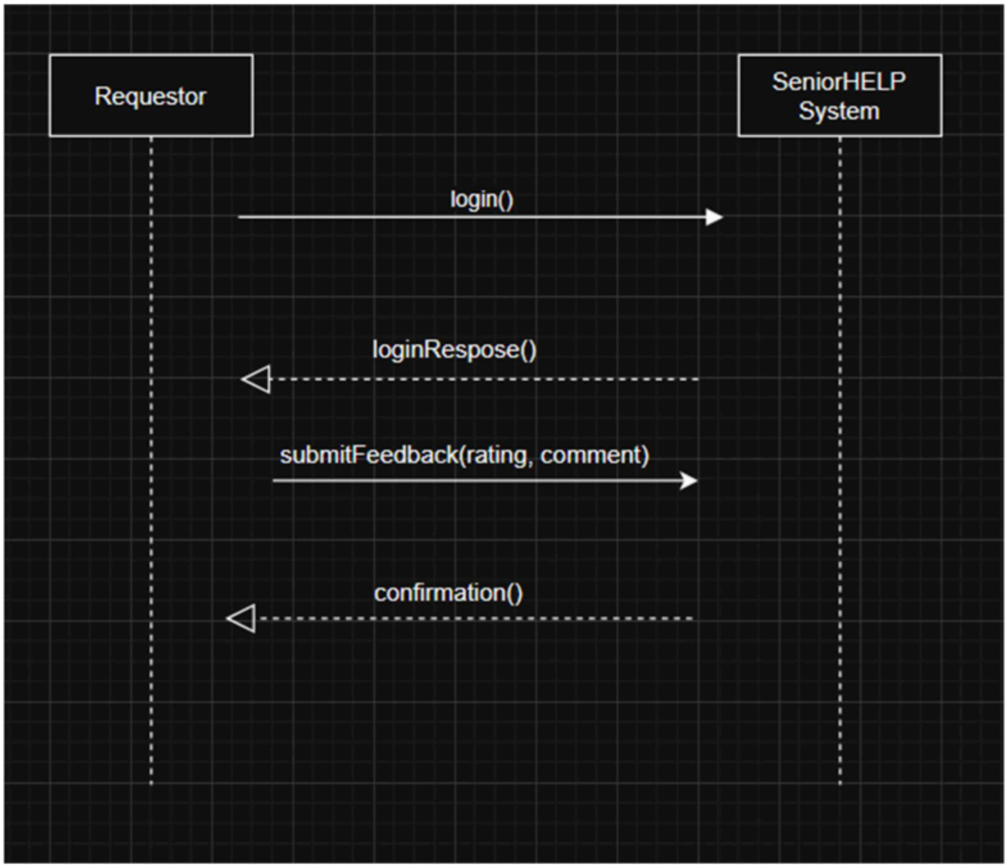
Technical Description:

This Diagram represents the Volunteer aspect of service.

- Actor: Service Provider (Volunteer).
- Core Messages: * viewRequests(): This request inquires about a list of pending tasks.
- acceptRequest(requestID): The volunteer agrees to take on a specific task.

- System Action: Executes a callback to updateRequestStatus("Assigned"). This stops two volunteers from taking the identical position.

Use Case 4: Accept Volunteer Request**(AADITYA CHAUDHARY(E2300548) - INDIVIDUAL TASK)**



Technical Description:

This SSD signifies the completion stage of the SeniorHELP service.

- The Requestor provides both qualitative and quantitative information via the submitFeedback() event.
- System Logic: The system employs a 'Black Box' method to change the request status to "Completed" by invoking a callback function. Once more, this serves to record the request in the history logs and to exclude it from the current pool of volunteers.

- Reputation Management: The input will influence the 'overall rating' of the Volunteer, aiding in maintaining the quality of the 'Home Volunteer' community.

Task 8:Use Case 1 Contracts

Use Case 1 Contracts – Register as Member

The Contracts outline the prerequisites, outcomes, and duties for the Register as Member scenario. This guarantees that the system operates reliably and accommodates additional scenarios such as soliciting volunteers.

Scenario: Sign Up as a Member

Main Actor: Requestor (Caregiver or Senior) / Volunteer

Interested Parties and Concerns:

- Requesters: Desire to register conveniently and safely.
- Volunteers: Interested in setting up an account to access opportunities for volunteering.
- System Administrators: Verify accurate user data and maintain system integrity.

Preconditions:

- The user is not yet registered in the system.
- The system is running and is able to take on new registrations.

Postconditions:

- A unique user ID is assigned when a new user account is established.
- The user is able to sign in and utilize relevant system functionalities.
- Administrator logs are refreshed with updated registration information.

Key Duties:

- Gather user details (name, contact information, address for elderly individuals).
- Verify user information (confirm that necessary fields are filled out and properly formatted).
- Save user data in the system's database.
- Inform the user of successful signup.
- Revise system records for administrative monitoring.

Additions / Inclusions:

- If any necessary data is absent, the system alerts the user to amend the details.
- If the user is already present, the system shows an error and offers password recovery or login options.

The system shows an error and recommends recovering the password or logging in. This agreement guarantees a reliable and safe registration process that other system features (such as asking for volunteers) can depend on

Use Case 2 Contract – Request Home Volunteer

Prepared by: **AAKROSHAN CHAUDHARY (Student ID: E2300551)**

Operation:
createVolunteerRequest(activityType, date, time, instructions)

Cross References

Use Case 2 – Request Home Volunteer

Preconditions:

- Requestor is logged in to the SeniorHELP system.
- Requestor's account status is active.

- Entered date and time are valid (i.e., not in the past).
- Required fields (activityType, date, time) have been provided.

Postconditions:

- A new object of type VolunteerRequest is created.
- volunteerRequest.requestID has been generated and assigned.
- volunteerRequest.activityType = activityType.
- volunteerRequest.date = date.
- volunteerRequest.time = time.
- volunteerRequest.instructions = instructions.
- volunteerRequest.status = “Available”.
- volunteerRequest is associated with the current Requestor.
- volunteerRequest is added to the system’s available request list.

Use Case 3 Contract – Accept Volunteer Request

Prepared by: **AADITYA CHAUDHARY (Student ID: E2300548)**

Operation:

acceptRequest(requestID)

Cross Reference:

Use Case 3 - Accept Volunteer Request

Preconditions:

- The Volunteer is logged into the SeniorHELP system.
- The Volunteer account is "Verified."
- A VolunteerRequest associated with the requestID exists.
- volunteerRequest.status = "Available."

Postconditions:

1. volunteerRequest.status = "Assigned."
2. The Volunteer is associated with the VolunteerRequest.
3. The VolunteerRequest is removed from the list of available requests.
4. The VolunteerRequest is added to the Volunteer's list of accepted requests.
5. Requestor is notified that the request is assigned.

Use Case 4 Contract – Provide Feedback

Prepared by: **BUDDHIMAN B.K (Student ID:(E2400004)**

Operation: ubmitFeedback(requestID, rating, comment)

Cross Reference: Use Case 4 – Provide Feedback

Preconditions:

1. The Requestor is logged into the SeniorHELP system.
2. A VolunteerRequest with the given requestID exists
3. volunteerRequest.status = "Assigned".
4. The volunteer service has been completed.

Postconditions:

1. volunteerRequest.status = "Completed".
2. A new Feedback instance is created.
3. feedback.rating = rating.
4. feedback.comment = comment.
5. The Feedback is associated with the VolunteerRequest.
6. The Volunteer's overall rating is updated.
7. The VolunteerRequest is archived from the active request list.

Conclusion

The development of the SeniorHELP system architecture can be considered a holistic effort towards addressing a critical social problem faced by seniors in Nepal. The systematic implementation of Object-Oriented Analysis and Design (OOAD) methodologies and principles helped us, Aakroshan Chaudhary, Aaditya Chaudhary, and Buddhiman B.K., to transform the user requirements into technical design artifacts.

The construction of detailed Activity Diagrams, Use Case Diagrams, and Analysis Class Diagrams helped us develop a solid structural foundation for the proposed system. The individual contributions towards developing System Sequence Diagrams (SSD) and Operation Contracts ensured that each and every event in the system is technically correct and logically consistent. The proposed system addresses the problem of seniors feeling isolated and alone and the absence of coordinated help by creating a safe, transparent, and easy-to-use system. The proposed system can be considered a viable system architecture that uses technology to promote community support and enhance the quality of life for seniors.

Refrences

Core Textbooks & UML Standards Larman, C. (2005). *Applying UML and patterns: An introduction to object-oriented analysis and design and iterative development* (3rd ed.). Prentice Hall.

Object Management Group. (2017). *OMG Unified Modeling Language (OMG UML) version 2.5.1*.

Sommerville, I. (2016). *Software Engineering* (10th ed.). Pearson.

Project Context (Nepal & Elderly Care) Central Bureau of Statistics. (2022). *National Population and Housing Census 2021: Age structure and demographic transition in Nepal*. Government of Nepal.

AI Tool Usage (Required for Policy Compliance) OpenAI. (2024). *ChatGPT (Feb 15 version) [Large language model]*.